

Probability and Statistics course

Introduction

Gabor Vigh

TTK Department of Probability Theory and Statistics
ELTE

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Source: Slides created based on [Baron, 2014]

Least squares estimators

Definition

Response or dependent variable (Y) is a variable of interest that we predict based on one or several predictors. **Predictors or independent variables** ($X^{(1)}, \dots, X^{(k)}$) are used to predict the values and behavior of the response variable Y .

Regression of Y on $X^{(1)}, \dots, X^{(k)}$ is the conditional expectation,

$$G(x^{(1)}, \dots, x^{(k)}) = E \left\{ Y \mid X^{(1)} = x^{(1)}, \dots, X^{(k)} = x^{(k)} \right\}.$$

It is a function of $x^{(1)}, \dots, x^{(k)}$ whose form can be estimated from data.

Definition

Residuals

$$e_i = y_i - \hat{y}_i$$

are differences between observed responses y_i and their fitted values $\hat{y}_i = \hat{G}(x_i)$.

Method of least squares finds a regression function $\hat{G}(x)$ that minimizes the sum of squared residuals

$$\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Linear regression

The **Linear regression model** assumes that the conditional expectation

$$G(x) = E\{Y \mid X = x\} = \beta_0 + \beta_1 x$$

is a linear function of x . As any linear function, it has an intercept β_0 and a slope β_1 . We minimize the sum of squared residuals

$$Q = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{G}(x_i))^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

We can do it by taking partial derivatives of Q , equating them to 0, and solving the resulting equations for β_0 and β_1 . The partial derivatives are

$$\frac{\partial Q}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i);$$

$$\frac{\partial Q}{\partial \beta_1} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i.$$

Linear regression

The Ordinary Least Squares (OLS) estimators for the intercept ($\hat{\beta}_0$) and the slope ($\hat{\beta}_1$) are:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1 = S_{xy} / S_{xx}$$

where S_{xx} and S_{xy} are the sums of squares: $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$, $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$

The total variation among observed responses is measured by the **Total Sum of Squares**:

$$SS_{TOT} = \sum_{i=1}^n (y_i - \bar{y})^2 = (n-1)s_y^2.$$

A portion of this total variation is attributed to predictor X and the regression model connecting predictor and response. This portion is measured by the **Regression Sum of Squares**:

$$SS_{REG} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = b_1^2 S_{xx}.$$

The rest of total variation is attributed to “error.” It is measured by the **Error Sum of Squares**:

$$SS_{ERR} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2.$$

This is the portion of total variation not explained by the model. It equals the sum of squared residuals that the method of least squares minimizes. Thus, applying this method, we minimize the error sum of squares.

$$SS_{TOT} = SS_{REG} + SS_{ERR}$$

Goodness of Fit

The goodness of fit, appropriateness of the predictor and the chosen regression model can be judged by the proportion of $SSTOT$ that the model can explain.

Definition

R -square, or **coefficient of determination** is the proportion of the total variation explained by the model, $R^2 = \frac{SS_{REG}}{SS_{TOT}}$.

It is always between 0 and 1, with high values generally suggesting a good fit.

Inference about the regression slope

Sampling distribution of a regression slope $\hat{\beta}_1$ is $\text{Normal}(\mu_{\hat{\beta}_1}, \sigma_{\hat{\beta}_1})$, where

$$\mu_{\hat{\beta}_1} = E(\hat{\beta}_1) = \beta_1$$

$$\sigma_{\hat{\beta}_1} = \text{Std}(\hat{\beta}_1) = \frac{\sigma}{\sqrt{S_{xx}}}$$

We estimate the standard error of $\hat{\beta}_1$ by

$$s(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}, \quad s = \sqrt{\frac{SS_{ERR}}{n-2}}$$

and therefore, use T -intervals and T -tests.

Following the general principles, a $(1 - \alpha)100\%$ confidence interval for the slope β_1 is

$$\text{Estimator} \pm t_{\alpha/2} \left(\begin{array}{c} \text{estimated} \\ \text{st. deviation} \\ \text{of the estimator} \end{array} \right) = \hat{\beta}_1 \pm t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}}.$$

Testing hypotheses $H_0 : \beta_1 = B$ about the regression slope, use the T -statistic $t = \frac{\hat{\beta}_1 - B}{s(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - B}{s/\sqrt{S_{xx}}}$. P-values,

acceptance and rejection regions are computed based on T -distribution with $n-2$ degrees of freedom.

Example

Example

A computer manager needs to know how the efficiency of her new computer program depends on the size of incoming data. Efficiency will be measured by the number of processed requests per hour. Applying the program to data sets of different sizes, she gets the following results:

Data size (gigabytes), x	6	7	7	8	10	10	15
Processed requests, y	40	55	50	41	17	26	16

In general, larger data sets require more computer time, and therefore, fewer requests are processed within 1 hour. The response variable here is the number of processed requests (y), and we attempt to predict it from the size of a data set (x).

Solution

We start by computing the following sample statistics:

$$n = 7, \quad \bar{x} = 9, \quad \bar{y} = 35, \quad S_{xx} = 56, \quad S_{xy} = -232, \quad S_{yy} = 1452.$$

Estimate the regression slope ($\hat{\beta}_1$) and intercept ($\hat{\beta}_0$):

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-232}{56} \approx -4.14$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 35 - (-4.14)(9) \approx 72.3.$$

Notice the negative slope. It means that increasing incoming data sets by 1 gigabyte, we expect to process 4.14 fewer requests per hour.

We have $SS_{TOT} = S_{yy} = 1452$ partitioned into $SS_{REG} = \hat{\beta}_1^2 S_{xx} = (-4.14)^2 \cdot 56 \approx 961$ and $SS_{ERR} = SS_{TOT} - SS_{REG} = 1452 - 961 = 491$.

R -square (coefficient of determination) is $R^2 = \frac{SS_{REG}}{SS_{TOT}} = \frac{961}{1452} \approx 0.662$. That is, 66.2% of the total variation of the number of processed requests is explained by the size of the data sets only.

Inference about the Slope: To test the null hypothesis $H_0 : \beta_1 = 0$, compute the T -statistic:

$$t = \frac{\hat{\beta}_1}{\sqrt{s^2/S_{xx}}} = \frac{-4.14}{\sqrt{98.2/56}} \approx -3.13.$$

Checking the T -distribution table with 5 degrees of freedom, we find that the P -value for the two-sided test is between 0.02 and 0.04. We conclude that the slope is moderately significant. Precisely, it is significant at any level $\alpha \geq 0.04$ and not significant at any $\alpha \leq 0.02$.

References



Baron, M. (2014).

Probability and statistics for computer scientists.

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